

MINE GASES

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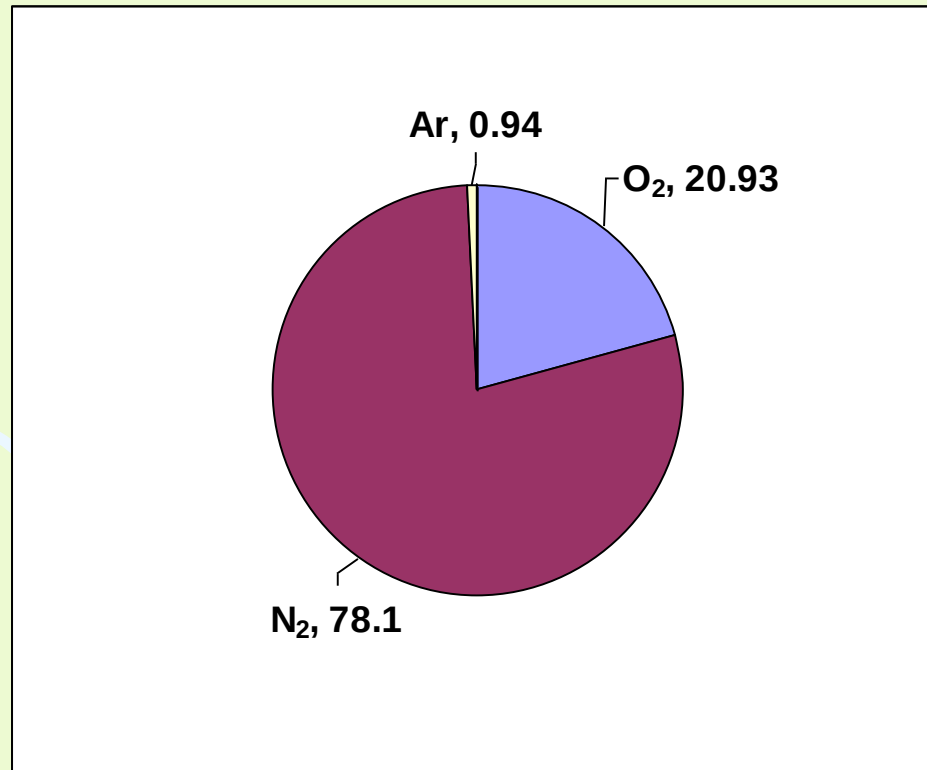
Atmospheric air and its composition

- The atmospheric air that we breath is a mixture of several gases.
- Its composition is practically constant over the whole surface of the earth from the sea level up to an altitude of at least 25 km.



Typical analysis of dry atmospheric air is given as below:

Constituent	% by volume
Oxygen	20.93
Nitrogen (including Ar and other rare inert gases)	79.04 (78.10 N ₂ + 0.94% Ar)
Carbon dioxide	0.03



- In addition to the above gases, the total amount of other gases is less than 0.01%.
- In normal atmospheric air, moisture is also always present, from a fraction of 1% up to a maximum of about 6%.



MINE GASES

In addition to O_2 and N_2 , the underground atmosphere contains small amounts of other gases, namely,

- Methane (CH_4),
- Carbon dioxide (CO_2),
- Carbon monoxide (CO),
- Unburned hydrocarbons,
- Partially oxidised hydrocarbons (aldehydes),
- Ammonia,
- Hydrogen sulphide (H_2S) and
- Sulphur dioxide (SO_2).

Composition of mine atmosphere

Analysis of air in the return airway of a gassy coal mine is generally as follows:

Constituent	% by volume
• Oxygen	20.28
• Nitrogen	78.90
• Carbon dioxide	0.36
• Methane	0.46
• It will be observed that return air is depleted in O ₂ content and contaminated by mine gases.	
• Impurities comes from exhalation by men, blasting, underground fires, burning of lights, bacterial action and gases given off from strata.	

- The foul or poisonous gases that pollute the air in coal mines are popularly known as '**damp**'.
- The word 'damp' originated from the German word 'dampf' means vapour, fog, fume or smog.

The common types of damp met with in mines are:

1. Blackdamp:

- It is a mechanical mixture of the extinctive gases, e. g. CO_2 and excessive N_2 .
- CO_2 may vary from 5-18% and rest is N_2 .
- Sometimes it is referred to as ***chokedamp*** or ***stythe***.
- The ill effect on health by blackdamp is due to lack of O_2 .

2. Firedamp:

- It is a mechanical mixture of inflammable gases emitted naturally from coal strata – mostly methane (CH_4) and air.
- Firedamp sometimes used synonymously with methane.
- It may be explosive depending on the composition (5.4-14.8% CH_4 in air).

3. Whitedamp:

- Mixture of CO and air.
- When a person dies his face becomes white and got its name as whitedamp.
- It is used synonymous with CO.

4. Stinkdamp:

- Mixture of H_2S and air.
- It is synonymous with sulphuretted hydrogen, H_2S (Hydrogen sulphide).
- H_2S is more poisonous than even CO.

5. Afterdamp:

- A mechanical mixture of gases existing in a mine after an explosion of firedamp or coal dust.
- Its composition is extremely variable but usually includes CO, CO₂, N₂ and sometimes H₂S, and SO₂ with **very small percentage of O₂**.
- Composition of afterdamp depends upon
 - initial composition of the mine atmosphere before an explosion and
 - how the combustion has taken place.
- A common afterdamp mixture consists of

O ₂	- 5%
N ₂	- 80-85%
CH ₄	- 0.3 to 0.5%
CO ₂	- 4-12% (generally 10%)

OXYGEN (O₂)

Properties of O₂

- Colourless, odourless and tasteless.
- Quite abundant on earth.
- Slightly soluble in water: 3% by volume at 20°C, i. e. 100 vol. of water dissolving only 3 vol. of O₂
- Its density is 1.428 kg/Nm³
- Slightly heavier than air: Specific gravity is 1.1, i.e. 1.1 times heavier than air
- It supports life and combustion.
- Essential in any atmosphere where men work.

Causes of O₂ deficiency in mines

In mines, the O₂ content in mine air decreases due to

- Breathing of men
- Burning of flame safety lamps
- Dilution of strata gases
- Oxidation of organic and inorganic substances e. g. coal, timber and pyrites
- Decay of timber by fungus growth/bacterial action
- Underground mine fires or explosion
- Appearance of other gases in higher concentrations e. g. CO and CH₄.

Table: O₂ Consumption at different work situation
(Haldane's work at NTP)

Nature of work	O ₂ consumed in lt/min
Rest in bed	0.237 lt/min.
Rest standing	0.328 lt/min.
Walking at 3.2 Km/hr	0.780 lt/min.
Walking at 6.4 Km/hr	1.595 lt/min.
Walking at 8.1 Km/hr or doing ordinary exercise	2.543 lt/min.
Extreme hard work	4.0 lt/min.

Requirements of O₂ in mine air as per mining laws:

- In India: Minimum 19%
- In USA: Minimum 19.5%
- In USSR Minimum 20%

PHYSIOLOGICAL EFFECTS OF O₂ DEFICIENCY

- Fall of O₂ from its usual level of 21% in the air to 18% does not represent any danger to human life.
- Human beings can survive in an atmosphere containing as low as 15% O₂.
- Canaries or small birds can survive if O₂% is as low as 8% and therefore, they are not good guides for sufficiency of O₂ for human beings.

Physiological effects of an O₂ deficient environment will vary from individual to individual and with the length of exposure

Conc. of O ₂ (% by volume)	Physiological effect
17	Faster and deeper breathing
15	Dizziness, buzzing in ears and rapid heartbeat
13	May lose consciousness with prolonged exposure
9	Fainting, unconsciousness
7	Life endangered
6	Convulsive movements and death

Methods of detecting O₂ deficiency:

Using flame safety lamp:

- Each 1% reduction in O₂% results in about 30% less light from oil lamps.
- Flame safety lamps extinguishes when O₂ content falls to about 17.5%.
- If safety lamp burns at a place it is safe for human beings to work there.

OXOR, an instrument for absorbing O₂ in Germany. It has a measuring accuracy of 0.2% O₂

Instrument based on fuel cells or electro chemical cells. This gives electrical signal for the presence of O₂

O₂ detector tubes are available in Poland, Russia etc.

NITROGEN

Properties of N₂

- Quite abundant in atmospheric air comprising about 4/5th of the atmosphere.
- Colourless, odourless and tasteless gas.
- It is nearly as heavy as air with Sp. Gravity 0.967.
- Practically insoluble in water: 100 vol. of water dissolving only 1.8 vol. of N₂ at 15 °C.
- It is inert: Neither burns nor support life or combustion
- It is important for the growth of plant and animal tissues.
- It does not undergo any alteration in the process of breathing.
- Nitrogen is sometimes used for quenching underground fires.

Occurrences of N₂

- Decomposition of organic substances
- Blasting with explosives – NG base
- No bad effect at high concentration – leads O₂ deff. with consequent physiological effect.

CARBON DIOXIDE (CO₂)

Physical Properties:

- It is a colourless, odourless gas and bitter in test (slight acid taste).
- It is one and half time heavier than air and has sp. gr. of 1.52.
- Highly soluble in water: 100 vol. of water dissolves at 15 °C 100 vol. of CO₂ forming a weak acid.
- It is not combustible and doesn't support life and combustion
- Small spaces in lungs absorb CO₂ – CO₂ in alveolar air is 5.6%. If it increases to 5.8% then breathing is doubled.

Physiological effect of CO₂

The effect is due to presence of CO₂, which is accompanied by corresponding depletion of O₂.

% of CO ₂ (by vol.)	Effect on man	Effect on lamp
0.03%	Normal	Normal
0.5%	Slight increase of depth and frequency of breathing	No effect
1%	Much deeper breathing	No effect
3%	Breathing rate is doubled, quick fatigue will result when working	It burns with difficulty
6%	Violent panting, rapid exhaustion	Extinguished
10%	Narcotic effect produced, unconsciousness if exposure is prolonged	
20%	Death within a few hours	

Sources of CO₂ in mines:

- Breathing of men (Avg. 50 litres per hour per worker).
- Burning of lamps (0.15 to 5 litres per minute per lamp).
- Decay of timbers.
- Slow oxidation of coal in coal mines (Spont. Combustion).
- Mine fires and explosions.
- Blasting.
- I.C. engines e. g. diesel locomotives.
- Emission from strata gas outbursts in coal and potash mines.
- Action of acid water on carbonates rocks.

Areas in the mine where CO₂ can be found:

- Dip side working at lower levels or layers
- Return air of the mine.
- Poorly ventilated areas.
- Sealed off areas of goaf.



Saturatory limit of CO₂

- India: Should not be more than 0.5%
 - U.K.: Should not be more than 0.2%
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CARBON MONOXIDE



- CO is produced whenever carbon or carbonaceous matter is burnt with insufficient supply of oxygen.
- It is the most dangerous gas encountered in coal and metal mines.

Properties of CO:

- (1) It is a colourless, odourless, tasteless and non-irritating gas.
- (2) It is slightly lighter than air (specific gravity : 0.967).
- (3) It is combustible but does not support combustion.
- (4) It burns in air at about 700°C with a light blue flame to CO_2 .
- (5) It is hardly soluble in water: 100 vols. of water dissolves only 2.3 vols. of gas at 20°C .
- (6) It forms an explosive mixture with air when present within the range of nearly 12 to 75% by vol.

OCCURRENCE:

The production of CO in a mine may be due to

1. Oxidation of coal and other carbonaceous matter:

- Incomplete oxidation may result in its formation
- Under normal mining conditions the percentage formed is negligible and harmless in the return air of a coal mine.

2. Spontaneous combustion of coal:

- This is the main source of production of dangerous % of CO in a coal mine.

3. Active mine fire:

- Active fire in underground mine forms CO in dangerous %.
- Timber fires can also produce CO

4. Methane or coal dust explosion:

- This is also one of the principal sources of CO.

- Gases produced by the explosions of CH_4 and coal dust invariably contain large % of CO which are responsible for deaths of victims.
- Coal dust explosions produce more CO.

5. Explosives/Blasting:

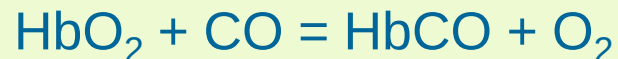
- Explosives contain the amount of O_2 required for complete chemical reaction but the chemical reaction when the explosive is blasted is seldom perfect and this results in the formation of CO.
- If the explosive is not oxygen balanced it produces CO.
- CO production is less if the explosive is blasted in strong rock than in weak rock

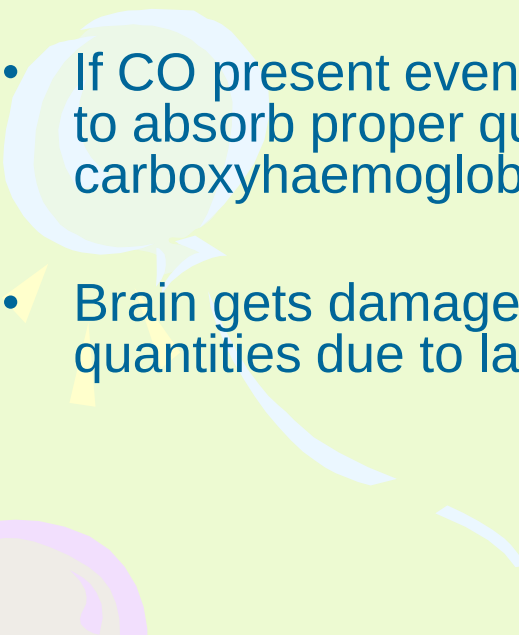
6. Underground machinery:

- Diesel equipments, air compressors, etc. when run faulty produce CO.
- Appreciable quantities of CO is produced in exhaust gas of internal combustion engines like diesel locomotives.
- In fact every machine produces some CO if proper lubrications are not used.

PHYSIOLOGICAL EFFECTS OF CO

- CO is a very poisonous gas and commonly referred to as the '**Silent Killer**'.
- It often affects the victims in such a manner that they fail to recognize the danger until it is too late.
- Its poisonous character is due to its great affinity for haemoglobin (oxygen-carrier) of the blood.
- Its affinity for haemoglobin is 300 times that of O₂.
- When CO reacts with haemoglobin it forms carboxyhaemoglobin (Hb.CO), a stable bright pink compound.
- The reaction between O₂, CO and haemoglobin is reversible and may be represented as



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- The rate at which CO combines with the haemoglobin depends on the
 - Exposure time
 - CO-concentration, and
 - The activity being performed.
 - If CO present even in small quantities in the inhaled air, it is difficult for blood to absorb proper quantity of O₂ to support life because of formation of carboxyhaemoglobin.
 - Brain gets damaged when CO breathed for a sufficiently long time in large quantities due to lack of O₂.
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Effects of breathing air containing CO on human beings

% of CO in air	Symptoms
0.02%	Headache, discomfort and possibility of collapse after 45 minutes at work or 2 hours at rest
0.12%	Palpitation/trembling after 10 minutes at work or 30 minutes at rest
0.2%	Unconsciousness after 10 minutes at work or 30 minutes at rest
0.5 to 1.0%	Death after 10-15 minutes of work

The after effects of CO poisoning are headache, loss of strength, and in some cases, even paralysis.

Detection and Estimation of CO

The following methods are used for detection and estimation of CO in mine atmospheres:

1. Using birds or mouse

- Exposing warm blooded birds or mice in cage to atmosphere containing gases,
- **Canaries** are used in U.S.A. and **Lal Muniahs** in India.
- Bird is a better indicator than mouse

2. Using colorimetric indicating detectors

- Based on the change in colour of the chemicals used for detection due to its reaction with CO

3. Using electronic measuring instruments (indicating and monitoring)

- Utilizes electronic sensors for detection of CO

1. Using birds or mouse

- Warm blooded birds or mouse are commonly used as they are affected much earlier than man by CO.
- On account of much faster breathing rates (about 20 times), the canaries are affected more quickly by the gas than a human being and thus give a warning signal for man to retreat to fresh air.
- Investigations carried out both in lab. and underground shown that at CO conc. greater than 0.18%, canaries are 3 times more sensitive than mice and 6 to 10 times more sensitive than man at rest.
- Such birds form the essential equipment of rescue team entering a mine after explosion or fire.
- Birds are taken in small cages (2 or 3 per cage) to place where CO is expected and their behaviour observed.

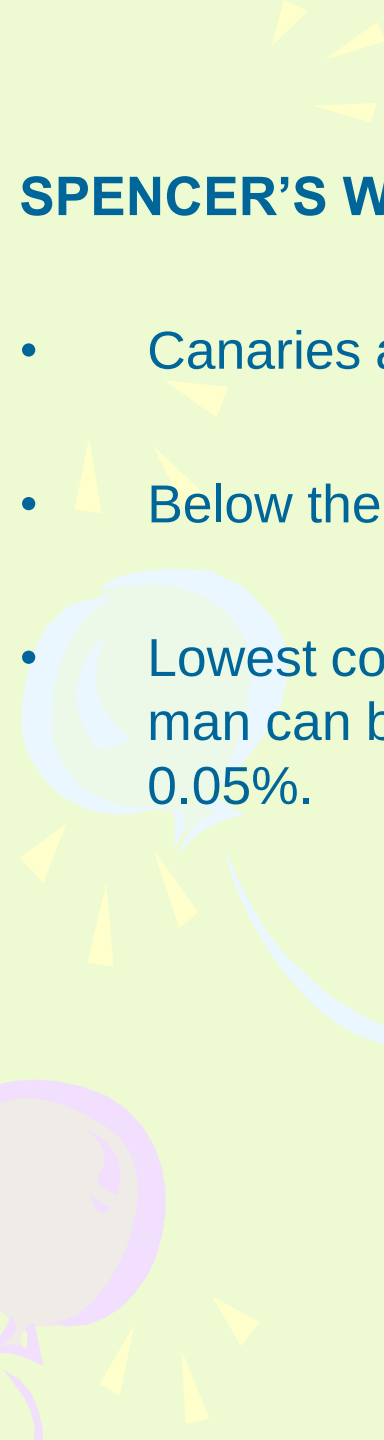
Effect of CO on muniah birds

- With 0.15% of CO present in the air a bird shows distress (ruffling of feathers, pronounced chirping and loss of liveliness) in 3 mts. and falls off its perch in 18 mts.
- With 0.3% CO the bird shows almost immediate distress and falls off its perch in 2-3 mts.
- Immediate signs of distress are not likely to be observed when exposed to only 0.1% of CO, they are visible only when the conc. is more than 0.3%.
- Ordinary sparrows are not suitable for CO detection and in an atmosphere containing CO to the extent of 1 to 2%, sparrows did not exhibit any sign of distress until they were dead.
- For atmospheres containing more than 0.15% CO, a cage with munia birds is a good indicator.

BURREL'S WORK :

Behaviour of Canaries exposed to CO

% CO	Effect
0.09%	Showed very slight distress after exposure for an hour
0.15%	Distress after 3 min and fall from perch after 8 min.
0.2%	They are distressed in one a half min and fall from the perch after 5 min.
0.29%	Fall from the perch in two and half minutes



SPENCER'S WORK:

- Canaries are useful when the CO concentration is more than 0.175%.
- Below the above concentration, canaries become less sensitive.
- Lowest concentration to make canaries insensitive is 0.125%, whereas, man can become unconscious with as low a concentration as 0.04 – 0.05%.

2. Using colorimetric indicating detectors

These detectors provide a rapid method of on-the-spot measurement of CO concentration.

Principle

- CO colorimetric detectors are based on the chemical reaction of CO with certain chemicals resulting in a change of colour.
- The chemicals are kept in detector tubes which are sealed at both the ends and can be stored for 1-2 years.
- The seals are broken at site just before detection of gas percentage and the tube inserted into an aspirator which draws in mine air.
- The air containing CO causes change in the colour of the chemicals in the tube.
- Estimation of CO is done on the basis of
 - matching changed colour with standard shades calibrated against the % of CO or
 - noting the length of tube over which the change of colour has occurred.

Some of the detectors are

The P. S. detector:

- This is manufactured by **Siebe Gorman and CO. Ltd., Britain.**
- It consists of a glass tube containing **silica gel** impregnated with light yellow **potassium palladium sulphite** with silica gel at both the ends to absorb other gases.
- A fixed volume of air (120cm^3) is drawn through the tube at a constant rate over a period of 2 min. through a calibrated orifice by operating a rubber aspirator.
- CO in the air turns **light yellow colour of Potassium-Palladium sulphite to brown.**
- The length of colour change from one end of the tube indicates the concentration of CO.
- The range of the instrument is 0.005-0.12% CO.

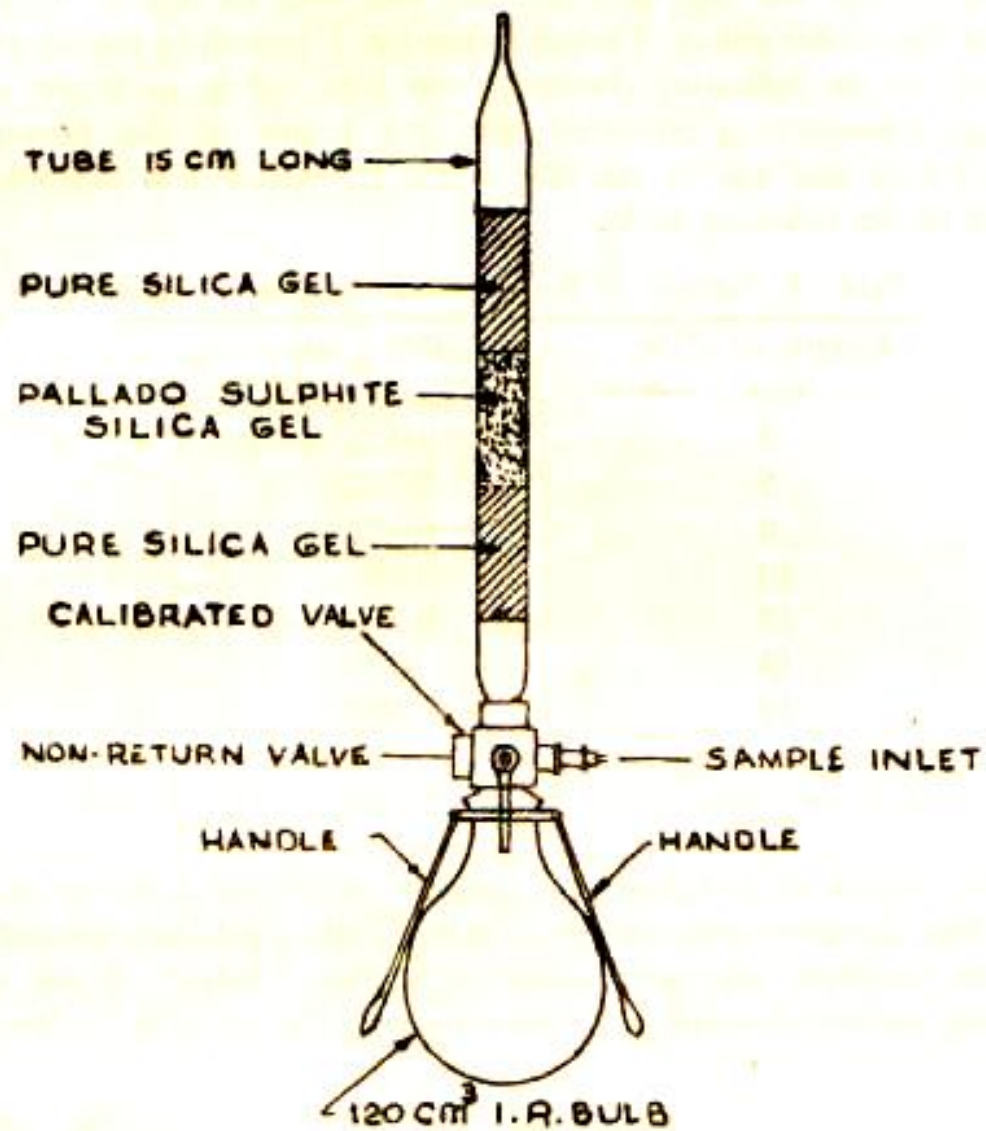


Fig. 65. Siebe Gorman P.S. Carbon Monoxide Detector.

B. The Hopcalite detector:

- Manufactured by the Mine Safety Appliances Co. USA.
- This detector utilizes the heat generated on oxidation of CO to CO₂.
- It consists of an **analyzing cell** containing **hopcalite** (a mixture of **manganese dioxide (MnO₂)** and **copper oxide (CuO)**).
- Air is drawn into the cell by a hand operated pump.
- CO in the air is catalytically oxidized to CO₂ by Hopcalite and generates heat.
- The rise in temp. so produced is recorded by a thermocouple connected in series to a meter which is calibrated in the % of CO.
- The instrument can read from 0.005 to 0.015 % CO.

C. The Hoolamite tube:

- Glass tube filled with **hoolamite** which is a mixture of **iodine pentoxide (I_2O_5)** and **sulphuric acid (H_2SO_4)**.
- A tube containing activated charcoal introduced between intake and hoolamite to prevent gases other than CO from reaching it.
- CO in the air due to reaction, reduces iodine pentoxide liberating iodine.
- The grayish white colour of the hoolamite in the tube turns green, brown or black depending upon the concentration of CO.
- Concentration of CO is estimated by comparing the colour with a standard colour chart provided with the instrument.

D. Carbon monoxide detector of M.S.A. Ltd.

- The detector consists of
 - an aspirator
 - a colour comparing scale/chart and
 - CO detecting tubes

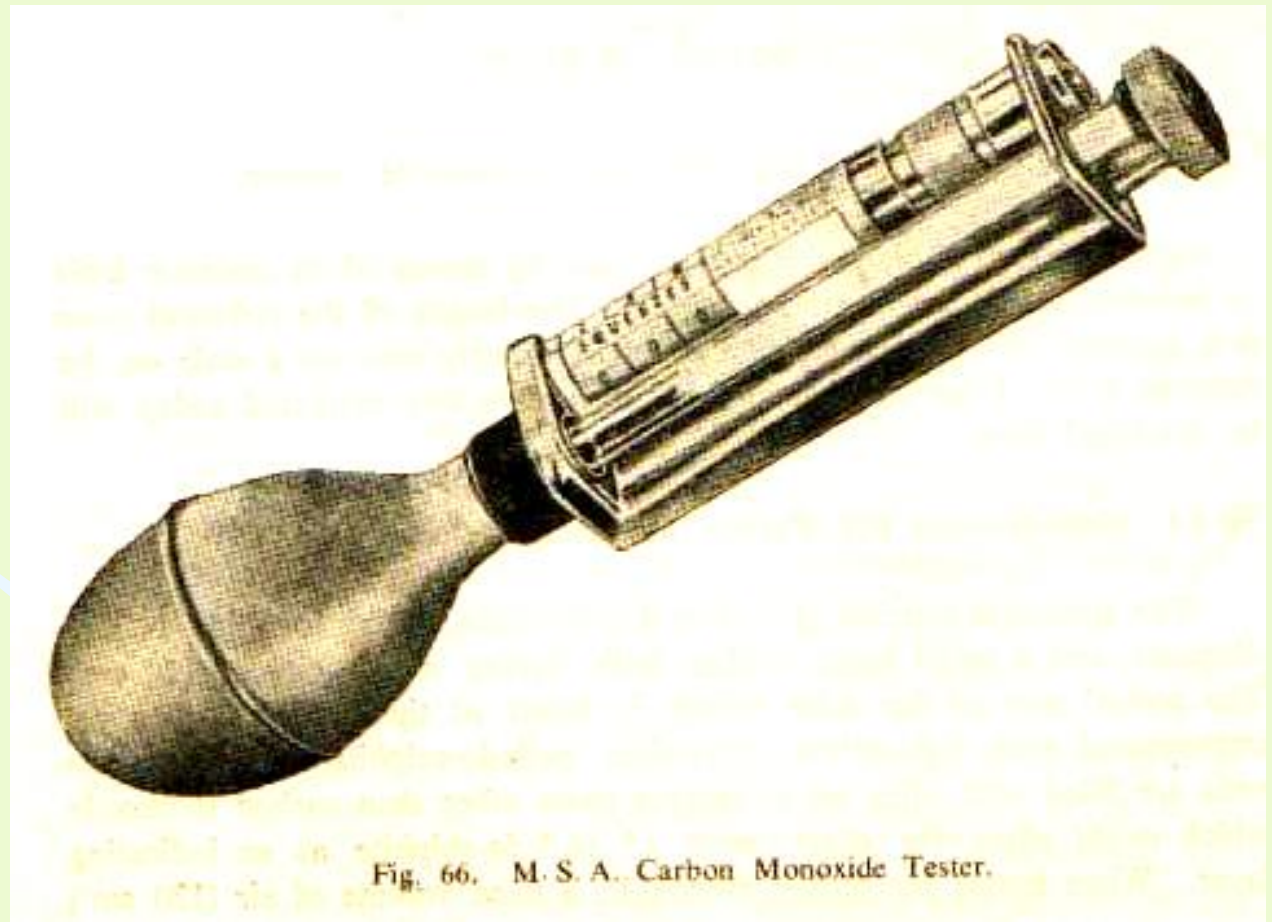


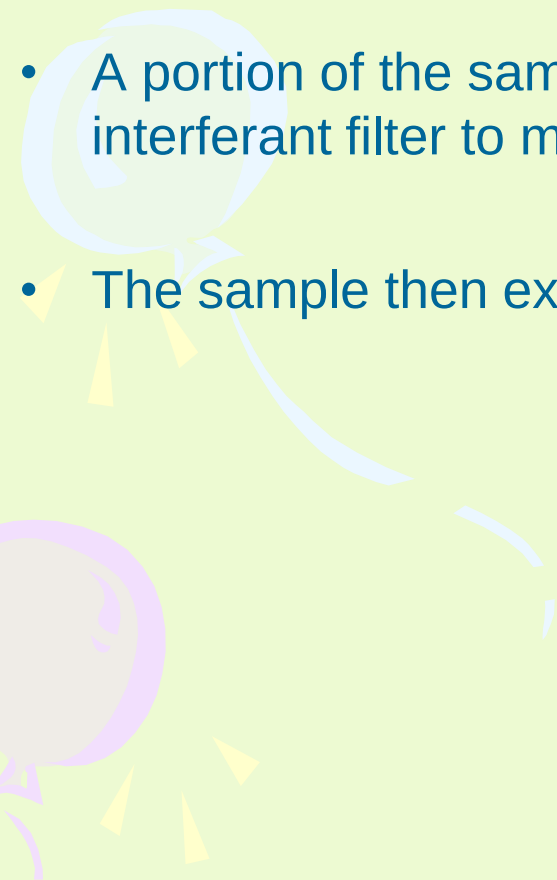
Fig. 66. M. S. A. Carbon Monoxide Tester.

- The aspirator (60 cc capacity) allows the user to use a variable no. of squeezes (1, 2 or 5), thereby enabling detection of CO conc. from as low as 10 ppm to as high as 1000 ppm by the colorimetric process.
- The tube contains **yellow silica gel** impregnated with **palladium sulphate (PdSO_4)** and **ammonium molybdate, $(\text{NH}_4)_2\text{MoO}_4$** and sealed at both the ends.
- The sealed tips are broken at the spot of detection and inserted into the CO detecting aspirator.
- As the aspirator bulb is squeezed, mine air is drawn into the tube through glass fibre and silica gel at the end of the tube so as to absorb gases other than CO.

- CO in air reacts with Ammonium Palladium Complex producing oxides, predominantly that of molybdenum and thereby change the yellow colour to shades of green depending upon the concentration of CO.
- The CO % is determined by matching with colour scale/chart provided with the instrument.
- The CO% read off corresponding to the no. of squeezes.
- Instrument is sensitive to a conc. of 0.01% CO and can detect a range of 0.01 to 0.4%.
- If the CO% is less, 5 squeezes of the aspirator bulb is necessary.

3. Using electronic measuring instruments

- For frequent measurement of CO, portable electronic measuring instruments are more accurate and economical than a detector with detector tubes.
- These are self-contained battery-powered instrument capable of measuring CO in air in a range of 0 to 100 or 0 to 500 ppm by vol.
- These are used for on-the-spot measurement of CO in air for fixed point continuous monitoring in any location.
- They utilize **electro-chemical polarographic cell sensors** and provide direct readings as well as audible and visual alarms if present at 50 ppm conc. or at any other desired level.

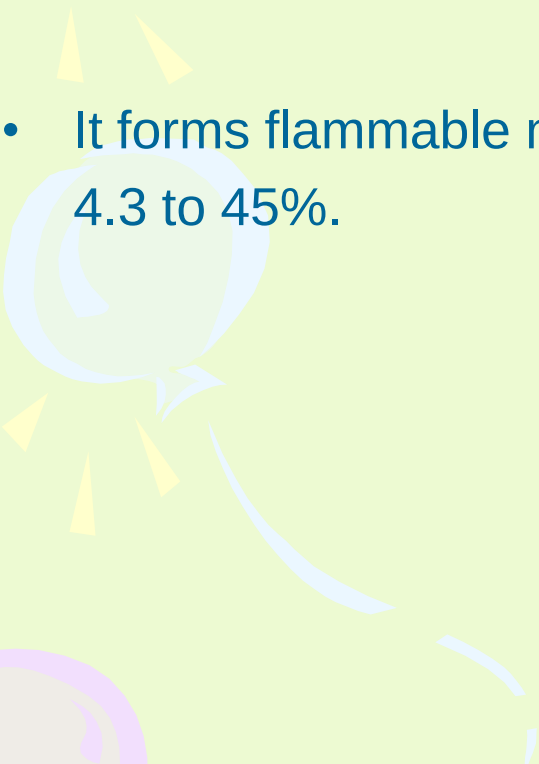
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- The rechargeable 2.4 V Ni-Cd battery pack sealed in a plastic case provides 8 to 10 hrs of sampling in fully charged condition.
 - A small pump draws an air sample into the inlet through a flowmeter and a factory-set flow control valve.
 - A portion of the sample is routed into the sensor sample chamber through an interferant filter to minimize their effects.
 - The sample then exhausts through a port in the sample chamber.
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HYDROGEN SULPHIDE

- H_2S was referred to as **stinkdamp** in early mining days.
- It is as poisonous as CO and may cause death in a short time if inhaled in large quantities.

Physical properties of H_2S :

- It is a colourless gas.
- Smells like rotten eggs.
- Slightly heavier than air. Has specific gravity 1.175 and density 1.5 kg/Nm^3 .
- It is readily soluble in water: 1 vol. of water dissolving about 3 vols. of H_2S .

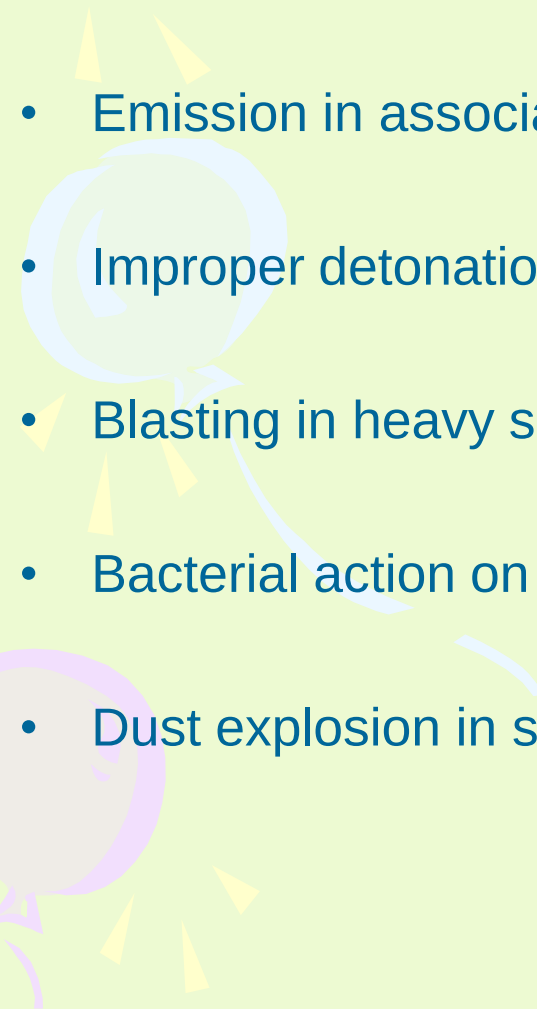
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- It is combustible but does not support combustion.
 - The gas burns in air with pale blue flame.
 - It forms flammable mixtures when mixed with air: the flammable range being 4.3 to 45%.
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Sources of H_2S in mine:

- H_2S gas occurs only in traces relatively in coal and metal mines.

H_2S is produced in mines from the following causes:

- Spontaneous heating of coal containing sulphur in the form of pyrites or finely disseminated organic compounds.
- Decay of organic matter especially timbers.
- Action of acid mine water on sulphides (iron pyrites).
- Methane blowers and outburst give small quantities of H_2S .

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- Emission from fissures and joints of strata especially in potash mines.
 - Emission from water of mineral springs when intersected.
 - Emission in association with firedamp.
 - Improper detonation of gun powder.
 - Blasting in heavy sulphide ore.
 - Bacterial action on sulphates.
 - Dust explosion in sulphide mines.
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Place of Occurrence of H₂S in Mines

- Found in stagnant water in old workings since highly soluble in water.
- Intermediate and main sump.
- In areas of gob fires or spontaneous heating.
- Old workings (decay of organic matter) due to poor ventilation.

PHYSIOLOGICAL EFFECTS OF H₂S

- H₂S is extremely toxic.
- Highly poisonous gas: it is more poisonous than CO but it gives an immediate warning.
- It causes irritation and inflammation of eyes and respiratory tracts at conc. of 50-100 ppm after 1 hr. of exposure.
- At high conc. of 200-700 ppm, it causes pain in throat and chest.

Physiological effects of H₂S

(Source: Forbes and Grove (1954) and Sunshine (1969))

Concentration	Symptoms
0.005 – 0.01%	Slight symptoms such as eye and respiratory tract irritation after 1 hr exposure
0.01%	Loss of odour after exposure of 15 minutes
0.02 – 0.07%	Increased eye irritation, headache, dizziness/sensation of whirling, nausea/vomiting, dryness, and pain in nose, throat, and chest
0.07 – 0.1%	Unconsciousness, cessation of respiration and death
0.1%	Death in a few minutes

DETECTION OF H₂S

BY SMELL:

- Presence of H₂S can be easily detected by the typical odour of rotten egg even at conc. as low as 0.0001% (1 ppm).
- This distinctive odour, however is noticed only in conc. less than 0.2%.
- At higher concentrations or following continued exposure to conc. as low as 0.01%, the gas paralyzes the sense of smell and the sensory response is not proportional to the conc. of the gas.
- Therefore, the sense of smell cannot be relied upon as a means of detection.

Using blotting paper:

- A blotting paper soaked in lead acetate changes its colour to black in presence of H₂S gas.
- A moist silver coin would also change to black in presence of H₂S due to formation of black sulphide on the surface.

Using colorimetric indicating detectors:

Using MSA H₂S detectors:

- MSA detector consists of **palladium chloride in an acetone-water solution**, sealed in a small glass tube which is surrounded with absorbent cotton.
- Presence of H₂S is indicated by colour change to dark in the cotton when the tube is crushed between thumbs and forefingers;
- Discolouration is compared with the scale on a standard colour chart supplied.

Using length-of-stain H₂S detectors:

- length-of-stain H₂S detectors using detector tubes are available from Draeger, MSA, Auer.
- The detector tube of MSA H₂S detector contains **white granules of activated aluminium oxide (alumina) coated with silver cyanide** which turn greyish-black on coming into contact with the H₂S in the air sample drawn through the tube due to formation of black silver sulphide.

- From the length of travel of discolouration through the granules, the percentage of H_2S in the air is directly read off the calibrated graduated scale.
- The instrument has a measuring range of 0 to 800 ppm.

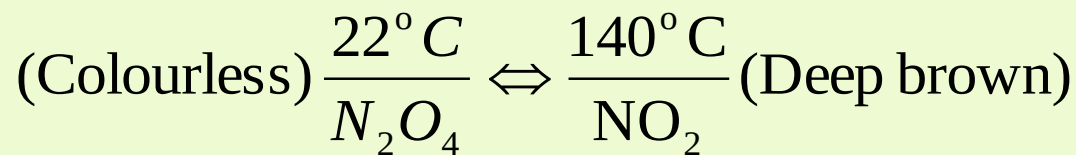
Portable H_2S detectors:

- Such detectors are manufactured by the MSA.
- Like portable CO indicators, these are self-contained battery-powered solid-state instruments using an electrochemical polarographic cell for their operation.
- They provide direct readings as well as audible and visual alarms if concentrations exceed present limits on the low and high ranges.
- The measuring ranges are 0 to 20 and 0 to 100 ppm H_2S in air with a min. detectable conc. of 0.5 ppm.
- A small pump draws a sample of air which is analyzed for H_2S .

NITROUS FUMES (OXIDES OF NITROGEN), NO_x

- N₂ is physiologically inert; however, under certain conditions it forms several oxides, some of which are extremely toxic.
- There are six oxides of N₂. These are NO, NO₂, N₂O, N₂O₃, N₂O₄, N₂O₅.
- N₂O₅ (Unstable white crystalline solid)
- NO (Nitric Oxide), NO₂ (Nitrogen dioxide), N₂O₄ (Nitrogen tetroxide) are having a biting and choking smell.
- They are yellow to reddish-brown in colour and have a smell of blasting powder fumes.
- They are easily dissolved by the moisture in mine air.
- **Nitrogen dioxide (NO₂) is the most toxic of these oxides of N₂.**

- Original product is NO, quickly combines with O₂ to form red fumes of nitrogen dioxide (NO₂) – this has got pungent smell like fuming nitric acid.
- As NO₂ cools down, it is slowly converted into colourless nitrogen tetroxide (N₂O₄)



- Out of nitrous fumes, NO₂ is most poisonous. NO is also poisonous, but less poisonous than NO₂.
- N₂O₅ is unstable crystalline solid – readily decomposes into O₂ and NO₂



SOURCES OF NITROUS FUMES MINES

- Nitrous fumes are formed during blasting by high explosives such as gelatine dynamite due to incomplete detonation.
 - **Explosives:** When explosives containing NG undergo combustion instead of detonations as may occur with weak detonators
- NG explosives with oxygen imbalance: an excess of O₂ produces nitrous fumes and an deficiency of O₂ produces CO.
- They tend to settle in the freshly blasted and dry muck pile, being about **1.5 times heavier than air** and released when muck pile is disturbed posing a hazard to the loader operator.
 - Nitrous fumes are also produced during the combustion process inside the cylinders of a diesel engine.
 - Diesel Loco: In small quantities in exhaust fumes of diesel loco.
 - Shafts and Tunnels: Shafts and tunnels where heavy shotfiring takes place in a confined space.

Physiological Properties:

- Nitrous fumes are highly poisonous and 7 to 10 times more toxic than CO.
- Nitrogen dioxide (NO₂) is 5 times more toxic than nitric oxide (NO).
- NO₂ combines with the hemoglobin of the blood about 300,000 times more readily than O₂ to form an extremely stable compound thereby quickly reducing the O₂-carrying capacity of the blood.
- Threshold limit value of NO₂ is given as 5 ppm as per DGMS circular.
- Even at low concs. (10-20 ppm) NO₂ exerts corrosive effect on lung tissues.
- As NO₂ is fairly soluble in water forming nitric acid, the moist air-passages of the nose, throat, and lungs are irritated immediately.
- Men affected by nitrous fumes show immediate symptoms of cough, nausea, choking, perspiration and headache, but later develop serious bronchial troubles such as bronchitis which may prove fatal within 48 hrs.

Table: Effects of Exposure to the Toxic Oxides of Nitrogen (Forbes and Grove (1954) and Christensen and Luginbyhl (1975))

Concentration ppm	Effect
5	Current TLV
60	Least amount causing immediate throat irritation
100	Least amount causing coughing
100-150	Dangerous even for short exposure
200-700	Rapidly fatal

Detection and estimation:

Using colour-stain detector tubes:

- Nitrous fumes can be detected by colour-stain detector tubes in conc. up to 100 ppm.
- The Draeger detector tubes measure nitrous fumes ($\text{NO} + \text{NO}_2$) in the ranges 0.5-10 ppm, 2-50 ppm, and 5-100 ppm.
- The detector tube contains a reddish-brown oxidation layer containing a **chromium compound** and a **yellow indicating layer of N-diphenyl benzidine** which changes colour to dark bluish-grey in the presence of the fumes.

Exposing a filter paper soaked with starch and Pottasium iodine solution to the fumes.

- Filter paper turns blue by the liberation of iodine in the presence of nitrous fumes.

Radon (Rn)

- The mining of uranium is different from any other form of mining because of the radioactive nature of the mineral and the release of radon gas and its progeny usually associated with the extraction of the mineral.
- Mine workers are exposed to radiation from the ore itself. And most importantly the inhalation of radon gas.
- Increased radiation exposure causes increased risk of cell damage.

RADON (Rn)

- Radon is a noble gas produced by the decay of radium 226 (^{226}Ra), so it can be breathed into the body.
- It is the only gas in the uranium decay series.
- It is a colourless, odourless, tasteless, partially soluble in water, non-flammable, and radioactive gas having a density of approx. 9.73 kg/m³ at 0°C and 100 kPa.
- It is chemically inert and not toxic, radioactive product of disintegration of radium.
- Found primarily in uranium mines, although present in trace amounts in other mines including coal mines.
- It releases the most harmful type of radiation-alpha particles. Has fairly short life 3.8 days.
- It is considered to be a health hazard due to its radioactivity and a great concern to ventilation engineers.
- **Breathing of high conc. of radon causes lung cancer.**

- 1st 4 decay daughters solid radioactive elements have a total half life less than 1 hour, 2 are alpha emitters & 2 are beta emitters.

- The next progeny lead-210 decays over 20 years.

- The progeny are highly reactive elements that can connect to airborne particles, such as aerosols (dust, smoke or moisture droplets) and then lodge in lung tissue.

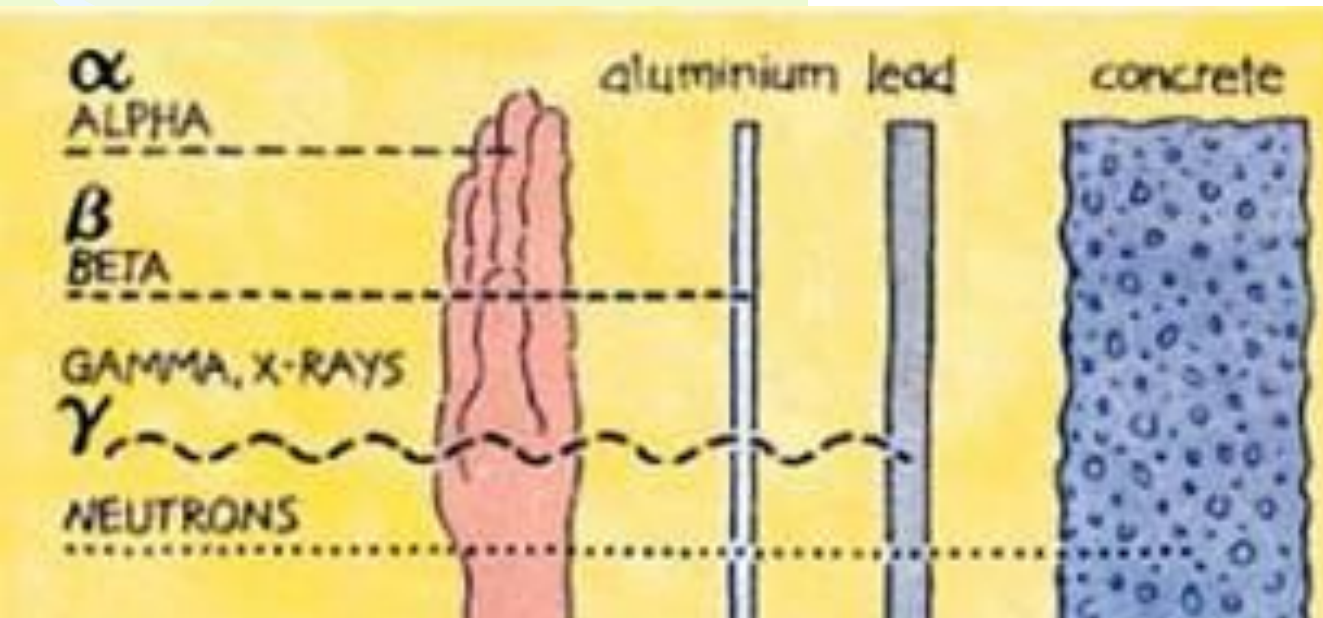
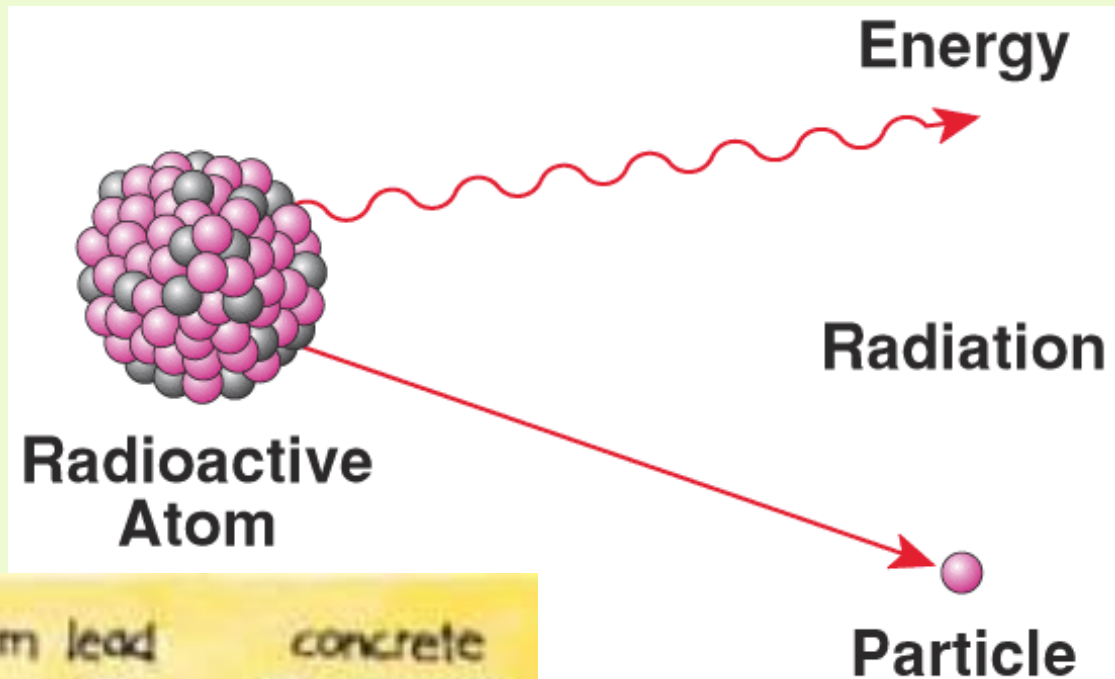
- Exposes sensitive lung tissue to radiation- hence lung cancer prevalence in mine workers.

URANIUM 238 (U238) RADIOACTIVE DECAY

type of radiation	nuclide	half-life
	 uranium-238	4.47 billion years
α	 thorium-234	24.1 days
β	 protactinium-234m	1.17 minutes
β	 uranium-234	245000 years
α	 thorium-230	8000 years
α	 radium-226	1600 years
α	 radon-222	3.823 days
α	 polonium-218	3.05 minutes
α	 lead-214	26.8 minutes
β	 bismuth-214	19.7 minutes
β	 polonium-214	0.000164 seconds
α	 lead-210	22.3 years
β	 bismuth-210	5.01 days
β	 polonium-210	138.4 days
α	 lead-206	stable

Types of ionising radiation

- Alpha, beta particles and gamma rays.



Radiation dose pathways for uranium workers

- External Gamma Radiation: from uranium ore
- Internal exposure to the lungs: via inhalation of ^{222}Rn gas and ^{222}Rn progeny.
- Internal exposure to the lungs and organs: via inhalation of airborne radioactive dust particles.

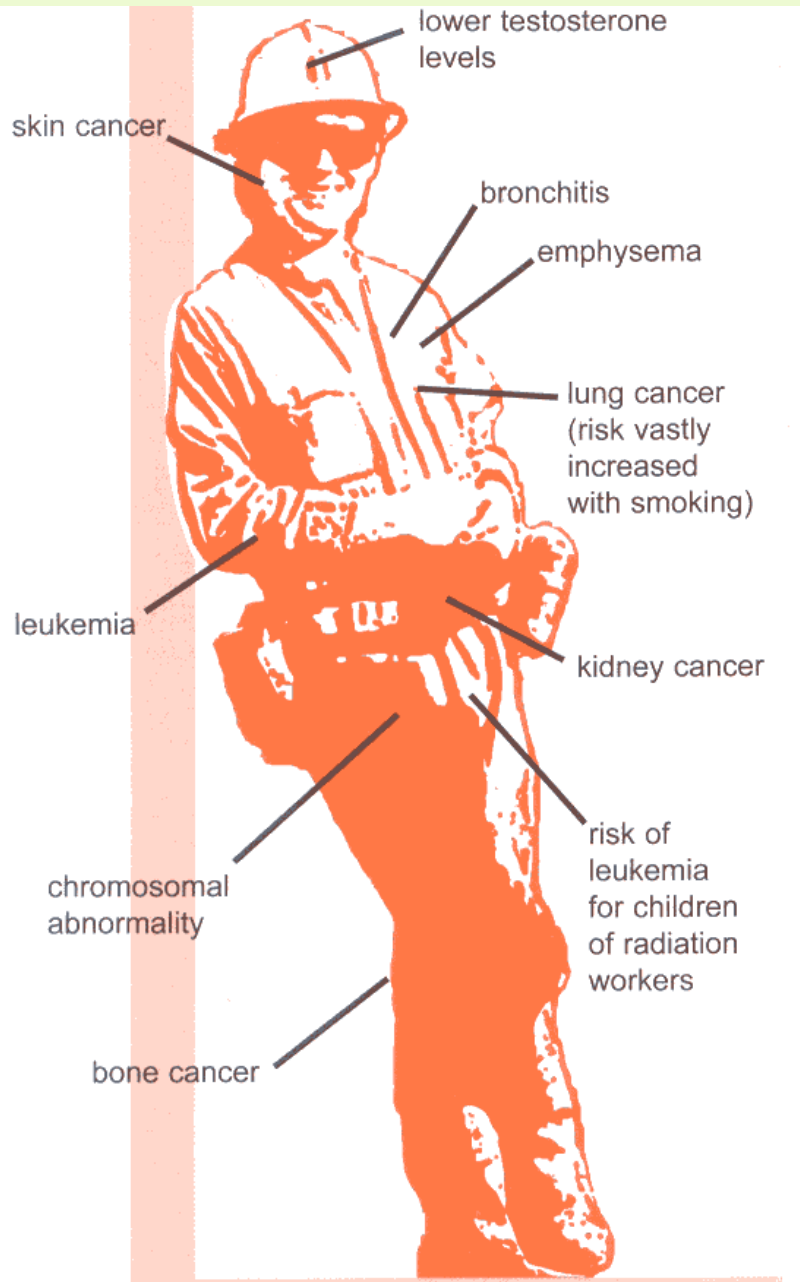
- Radon diffuses from the rock strata into the mine environment where the decay process continues with the formation of radium A, which decays to radium B, which produces radium C, and so forth.
- During the decay process, often alpha or beta particles and gamma rays are emitted.
- Alpha particles are positively charged, have low energy and can travel 3.8 to 7.6 cm in air.
- They do not penetrate the outer layers of skin and can be stopped by heavy clothing.
- Beta particles are negatively charged and can penetrate skin and a few mm of lead.
- Gamma radiation is electromagnetic radiation and has a relatively high penetration.

- Radon has no stable isotope and has atomic masses ranging between 204 and 224.
- Radon 222 (^{222}Rn) is the long lived isotope with a half-life of 3.82 days.
- All of the radon isotopes undergo radioactive decay producing their decay products (known as daughter products) until equilibrium is reached when (^{206}Pb) is formed.
- Radon daughter products are atoms of solid matter having relatively short half-lives.
- In the disintegration process, the uranium-238 decays to lead-206.
- The half-life of a radioactive substance is the time required for a given amount of that substance to lose one half of its radioactivity.
- For example, the half life of uranium-238 is approx. 4.5 billion years; radium, 1622 years; and radon 3.82 days.

Sources of Radon in Mines

- Radon is found in small concentrations in most hard rock mines.
- Coals are also known to contain traces of radium which can produce radon in underground mines.
- In uranium mines the radiation level is quite high and directly prop. to the grade of the uranium ore.
- The radon gas diffuses out into mine atmosphere through cracks and fissures or by way of mine water which dissolves radon.
- The emanation is enhanced if the host rock has a weak or porous structure.

Health risks on miners



- Increased risk of cancers.
- Significant reduction in testosterone levels.
- Increases in chromosome aberrations.
- when miners father children, increasing the risks of genetic abnormalities and leukaemia in their offspring.
- Other adverse health effects – (such as strokes and heart diseases) are associated with high doses.

Limiting ionizing dose level

- The ionizing electromagnetic radiation dose is commonly measured in mSv (milliSievert, the SI unit).
- $1 \text{ Sv} = 1 \text{ J/kg}$.
- Since 1934, ICRP (International Commission on Radiological Protection) has been setting the limiting ionizing dose for mine workers.
- In 1934 the limiting dose was 300 mSv/year, but was changed to 150 in 1956 and to 20 mSv/year in 1990.

Reduction of conc. of radon and daughter products in mine environment

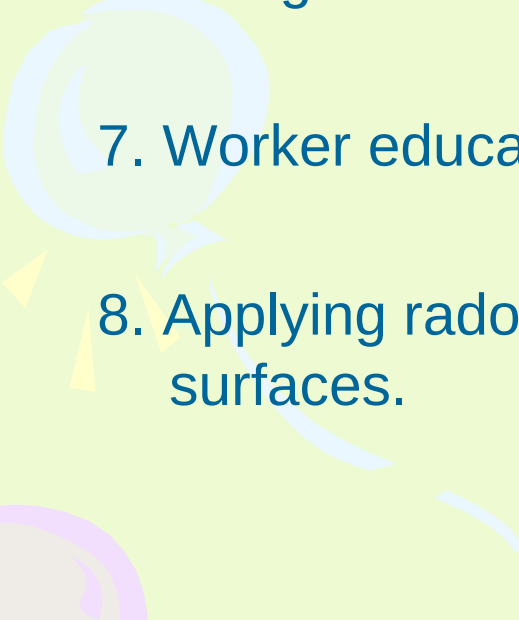
Conc. of radon and daughter products in the mine environment is controlled by

1. Providing adequate quantity of fresh air containing as low a concentration of radon as possible to the working section.
2. Providing high air velocities to keep the air residence time to a minimum.
3. Avoiding local recirculation of air.
4. Selecting a suitable mining method.



5. Sealing-off worked out sections effectively.

6. Using air-stream helmets for selected tasks.



7. Worker education.

8. Applying radon barrier sealant coatings to exposed rock surfaces.

